

234 quiz 5 material

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March 2026

1 Wave Equation

1.1 Dirichlet-Dirichlet

We have

$$\begin{cases} c^2 u_{xx} = u_{tt} \\ u(x, 0) = g(x) \\ u(0, t) = u(L, t) = 0 \end{cases}$$

Then applying separation of variables yields the following eigenvalue problems:

$$\begin{cases} X'' + \lambda X = 0 & X(0) = X(L) = 0 \\ T'' + c^2 \lambda T = 0 & T(0) = g(x) \end{cases}$$

The first eigenvalue problem has solution:

$$\lambda = \left(\frac{n\pi}{L}\right)^2 \quad X_n(x) = \sin\left(\frac{n\pi}{L}x\right)$$

The second problem has the solution:

$$T_n(t) = B_n \cos\left(\frac{n\pi c}{L}t\right) \quad B_n = \frac{2}{L} \int_0^L g(x) \sin\left(\frac{n\pi}{L}x\right) dx$$

$$\Rightarrow u(x, t) = \sum_{n=1}^{\infty} \left(\frac{2}{L} \int_0^L g(x) \sin\left(\frac{n\pi}{L}x\right) dx \right) \sin\left(\frac{n\pi}{L}x\right) \cos\left(\frac{n\pi c}{L}t\right)$$

1.2 Neumann-Neumann

We have

$$\begin{cases} c^2 u_{xx} = u_{tt} \\ u(x, 0) = g(x) \\ u_x(0, t) = u_x(L, t) = 0 \end{cases}$$

Applying separation of variables yields

$$\begin{cases} X'' + \lambda X = 0 & X'(0) = X'(L) = 0 \\ T'' + c^2 \lambda T = 0 & T(0) = g(x) \end{cases}$$

Solution of the eigenvalue problems are the following:

$$\lambda = \left(\frac{n\pi}{L}\right)^2 \quad X_n(x) = \cos\left(\frac{n\pi}{L}x\right)$$

Additionally, if $n = 0$, Neumann condition allows $X = 1$. Thus, we must include a steady-state term!

$$T_n(t) = B_n \cos\left(\frac{n\pi c}{L}t\right) \quad B_n = \frac{2}{L} \int_0^L g(x) \cos\left(\frac{n\pi}{L}x\right) dx$$

For $n = 0$ this yields the solution $T_n(t) = A_0 + B_0 t$ but due to homogeneity of the boundary condition, $B_0 = 0$. By substituting in the initial condition we get:

$$g(x) = A_0 + \sum_{n=1}^{\infty} B_n \cos \frac{n\pi}{L}x \implies A_0 = \frac{1}{L} \int_0^L g(x) dx, \quad B_n = \frac{2}{L} \int_0^L g(x) \cos\left(\frac{n\pi x}{L}\right) dx$$

Thus, the Neumann-Neumann problem has the solution

$$\implies u(x, t) = \frac{1}{L} \int_0^L g(x) dx + \sum_{n=1}^{\infty} \left(\frac{2}{L} \int_0^L g(x) \cos\left(\frac{n\pi}{L}x\right) dx \right) \cos\left(\frac{n\pi c}{L}t\right) \cos\left(\frac{n\pi}{L}x\right)$$

If boundary condition was not homogeneous we would have to find the value of B_0 and the solution would have a linear component outside the summation as well!

2 Laplace's Equation in Cartesian

2.1 Dirichlet-Dirichlet With Varying x Endpoint

We have

$$\begin{cases} u_{xx} + u_{yy} = 0 \\ u(0, y) = 0, u(L, y) = f(y) \\ u(x, 0) = u(x, H) = 0 \end{cases}$$

Separation of variables yields the following eigenvalue problems

$$\begin{cases} X'' - \lambda X = 0 & X(0) = 0, X(L) = f(y) \\ Y'' + \lambda Y = 0 & Y(0) = Y(H) = 0 \end{cases}$$

The second eigenvalue problem has the solution

$$\lambda = \left(\frac{n\pi}{H}\right)^2 \quad Y_n(y) = \sin\left(\frac{n\pi}{H}y\right)$$

The first eigenvalue problem then has the solution

$$X_n = A_n \sinh\left(\frac{n\pi}{H}x\right) \implies A_n \sinh\left(\frac{n\pi L}{H}\right) = \frac{2}{H} \int_0^H f(y) \sin\left(\frac{n\pi}{H}y\right) dy$$

The coefficient is determined by plugging in $x = L$ in the fourier series solution and then using the fourier coefficient formula. Thus, the general solution to the equation is given by the series

$$\implies u(x, y) = \sum_{n=1}^{\infty} \left(\frac{2}{H \sinh(n\pi L/H)} \int_0^H f(y) \sin\left(\frac{n\pi}{H}y\right) dy \right) \sinh\left(\frac{n\pi}{H}x\right) \sin\left(\frac{n\pi}{H}y\right)$$

2.2 Dirichlet-Dirichlet With Varying y Endpoint

We have

$$\begin{cases} u_{xx} + u_{yy} = 0 \\ u(0, y) = u(L, y) = 0 \\ u(x, 0) = 0, u(x, H) = g(x) \end{cases}$$

Separation of variables yields the following eigenvalue problems

$$\begin{cases} X'' - \lambda X = 0 & X(0) = X(L) = 0 \\ Y'' + \lambda Y = 0 & Y(0) = 0, Y(H) = g(x) \end{cases}$$

The solution to the first eigenvalue problem is

$$\lambda = \left(\frac{n\pi}{L}\right)^2 \quad X_n(x) = \sin\left(\frac{n\pi}{L}x\right)$$

For the second eigenvalue problem we then have

$$Y_n = A_n \sinh\left(\frac{n\pi}{L}y\right) \implies A_n \sinh\left(\frac{n\pi H}{L}\right) = \frac{2}{L} \int_0^L g(x) \sin\left(\frac{n\pi}{L}x\right) dx$$

The coefficient is determined by plugging in $x = L$ in the fourier series solution and then using the fourier coefficient formula. Thus, the general solution to the equation is given by the series

$$\implies u(x, y) = \sum_{n=1}^{\infty} \left(\frac{2}{L \sinh(n\pi H/L)} \int_0^L g(x) \sin\left(\frac{n\pi}{L}x\right) dx \right) \sinh\left(\frac{n\pi}{L}y\right) \sin\left(\frac{n\pi}{L}x\right)$$

2.3 Neumann-Neumann Homogeneous Problem

We have

$$\begin{cases} u_{xx} + u_{yy} = 0 \\ u_x(0, y) = 0, u_x(L, y) = f(y) \\ u_y(x, 0) = u_y(x, H) = 0 \end{cases}$$

Separation of variables yields the following eigenvalue problems

$$\begin{cases} X'' - \lambda X = 0 & X'(0) = 0, X'(L) = f(y) \\ Y'' + \lambda Y = 0 & Y'(0) = Y'(H) = 0 \end{cases}$$

The second eigenvalue problem has the solution

$$\lambda = \left(\frac{n\pi}{H}\right)^2 \quad Y_n(y) = \cos\left(\frac{n\pi}{H}y\right)$$

Then the first eigenvalue problem has the solution

$$X_n(x) = A_0 + A_n \cosh\left(\frac{n\pi}{H}x\right) \implies A_n \frac{n\pi}{H} \sinh\left(\frac{n\pi L}{H}\right) = \frac{2}{H} \int_0^H f(y) \cos\left(\frac{n\pi}{H}y\right) dy$$

Using the non-homogeneous boundary condition, we can match

$$A_0 = \frac{1}{H} \int_0^H f(y) dy$$

Thus, the full solution to a homogeneous Neumann problem is

$$\implies u(x, y) = \frac{1}{H} \int_0^H f(y) dy + \sum_{n=1}^{\infty} \left(\frac{2}{n\pi \sin(n\pi L/H)} \int_0^H f(y) \cos\left(\frac{n\pi}{H}y\right) dy \right) \cosh\left(\frac{n\pi}{H}x\right) \cos\left(\frac{n\pi}{H}y\right)$$

A similar approach is taken for the y direction boundary depending on some function $g(x)$.

2.4 Derivation of Laplace Equation in Polar Coordinates

We know the metric tensor

$$g_{mn} = \begin{pmatrix} 1 & 0 \\ 0 & r^2 \end{pmatrix} \implies g = r^2, \quad g^{mn} = \begin{pmatrix} 1 & 0 \\ 0 & \frac{1}{r^2} \end{pmatrix}$$

We also know the laplace operator in general coordinates:

$$\Delta u = \frac{1}{\sqrt{g}} \partial_m (\sqrt{g} g^{mn} \partial_n u)$$

Using polar coordinates then,

$$\Delta u = \frac{1}{r} \partial_r (r u_r) + \frac{1}{r} \partial_\theta \left(r \cdot \frac{1}{r^2} u_\theta \right) = u_{rr} + \frac{1}{r} u_r + \frac{1}{r^2} u_{\theta\theta}$$

Thus, Laplace's equation in polar coordinates has the form:

$$\Delta u = u_{rr} + \frac{1}{r} u_r + u_{\theta\theta} = 0$$

The separation of variables is the following:

$$\begin{aligned}
u(r, \theta) = R(r)\Phi(\theta) &\implies R''\Phi + \frac{1}{r}R'\Phi + \frac{1}{r^2}R\Phi'' = 0 \\
&\implies \frac{r^2R'' + rR'}{R} = \frac{-\Phi}{\Phi} = \lambda \\
&\implies r^2R'' + rR' - \lambda R = 0, \quad \Phi' + \lambda\Phi = 0
\end{aligned}$$

2.5 Laplace Equation Inside a disk

For a disc of radius $r = \rho$, with homogeneous condition on the boundary, and a periodicity:

$$\begin{cases} u_{rr} + \frac{1}{r}u_r + \frac{1}{r^2}u_{\theta\theta} = 0 \\ u(\rho, \theta) = f(\theta) \\ \theta \in [-\pi, \pi] \end{cases}$$

Since the points $u(r, \pi)$ and $u(r, -\pi)$ are identical, we impose the boundary condition $\Phi(\theta) = \Phi(\theta + 2\pi)$, $\Phi'(\theta) = \Phi'(\theta + 2\pi)$. Hence,

$$\begin{cases} r^2R'' + rR' - \lambda R = 0 & R(\rho) = f(\theta) \\ \Phi' + \lambda\Phi = 0 & \Phi(-\pi) = \Phi(\pi), \Phi'(-\pi) = \Phi'(\pi) \end{cases}$$

For the second equation, we have:

$$\lambda = n^2 \quad \Phi_n(\theta) = A_n \cos(n\theta) + B_n \sin(n\theta)$$

If $\lambda = 0$, the second equation becomes:

$$R_0 = c_1 + c_2 \ln r, \quad ||u(0, \theta)|| < \infty \implies c_2 = 0$$

Suppose $c_1 = A_0$, then for $\lambda = n^2$:

$$r^2 R'' + r R' - n^2 R = 0 \implies R(r) = C_1 r^n + C_2 r^{-n}$$

Since we need $\|u\| < \infty$, and since we know $r \geq 0$, then $C_2 = 0$. Hence,

$$R(r) = r^n$$

Now, since $R(\rho) = f(\theta)$ we get:

$$f(\theta) = A_0 + \sum_{n=1}^{\infty} \rho^n A_n \cos(n\theta) + \rho^n B_n \sin(n\theta)$$

which means

$$A_0 = \frac{1}{2\pi} \int_{-\pi}^{\pi} f(\theta) d\theta \quad A_n = \frac{1}{\pi \rho^n} \int_0^{\pi} f(\theta) \cos(n\theta) d\theta \quad B_n = \frac{1}{\pi \rho^n} \int_0^{\pi} f(\theta) \sin(n\theta) d\theta$$

Hence:

$$\implies u(r, \theta) = A_0 + \sum_{n=1}^{\infty} r^n [A_n \cos(n\theta) + B_n \sin(n\theta)]$$